# Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT | JEE-MAIN-Shift-2 | Date: 24-03-2026

Time: 03.00Pm to 06.00Pm | GTM-27 | Max. Marks: 300

## KEY SHEET

## MATHEMATICS

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | **2** | 2 | **2** | 3 | **1** | 4 | **1** | 5 | **4** |
| 6 | **1** | 7 | **2** | 8 | **3** | 9 | **3** | 10 | **2** |
| 11 | **1** | 12 | **2** | 13 | **2** | 14 | **1** | 15 | **4** |
| 16 | **2** | 17 | **2** | 18 | **2** | 19 | **1** | 20 | **3** |
| 21 | **2** | 22 | **25** | 23 | **32** | 24 | **103** | 25 | **7** |

## PHYSICS

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|---|---|---|---|---|---|---|---|---|---|
| 26 | **3** | 27 | **1** | 28 | **2** | 29 | **2** | 30 | **2** |
| 31 | **2** | 32 | **4** | 33 | **1** | 34 | **4** | 35 | **1** |
| 36 | **4** | 37 | **1** | 38 | **1** | 39 | **4** | 40 | **4** |
| 41 | **3** | 42 | **1** | 43 | **4** | 44 | **1** | 45 | **3** |
| 46 | **4** | 47 | **3** | 48 | **4** | 49 | **20** | 50 | **6** |

## CHEMISTRY

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 51 | **1** | 52 | **1** | 53 | **1** | 54 | **2** | 55 | **1** |
| 56 | **4** | 57 | **2** | 58 | **2** | 59 | **1** | 60 | **3** |
| 61 | **2** | 62 | **2** | 63 | **2** | 64 | **2** | 65 | **1** |
| 66 | **4** | 67 | **3** | 68 | **2** | 69 | **4** | 70 | **1** |
| 71 | **2** | 72 | **2** | 73 | **3** | 74 | **4** | 75 | **4** |



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# SOLUTION

# MATHEMATICS

1. $S=\left(1+\frac{1}{2}+\frac{1}{3}+...+\frac{1}{m}\right)+2\left(\frac{1}{2}+\frac{1}{3}+....+\frac{1}{m}\right)+3\left(\frac{1}{3}+\frac{1}{4}+....+\frac{1}{m}\right)+....$

$+(m-1)\left(\frac{1}{m-1}+\frac{1}{m}\right)+m\left(\frac{1}{m}\right)=(1)(1)+\frac{1}{2}(1+2)+\frac{1}{3}(1+2+3)+\frac{1}{4}(1+2+3+4)$

$+....+\frac{1}{m}(1+2+3+.....+m)=\sum_{k=1}^{m}\frac{1}{k}.\frac{k(k+1)}{2}=\frac{1}{2}\sum_{k=1}^{m}(k+1)$

$=\frac{m}{4}[2+(m+1)]=\frac{1}{4}m(m+3)$

2. (P) Let 1 be the image of first $x_1$ elements of set A and 2 be the image of next $x_2$ elements of set A and so on

$\Rightarrow x_1+x_2+x_3+x_4=6$; for onto function $x_1\geq 1$

$\Rightarrow$ Number of onto function = coefficient of $x^6$ in $(x+x^2+x^3+....)^4$

= coefficient of $x^2$ in $(1-x)^4 = {}^{4-2-1}C_2=10$

$\Rightarrow$ Total number of onto function (non – increasing or non – decreasing) = 20

Agent total number of into and onto function (non – increasing or non – decreasing)

$=\left(2\times{}^{4-6-1}C_6-4\right)=164$

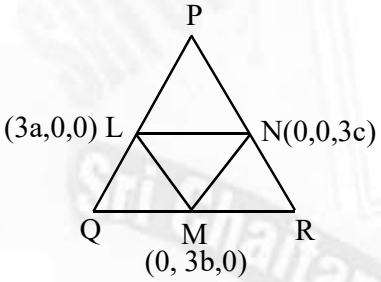
(R) As f(i) = i $\Rightarrow$ each element has 3 choices i.e. total number of ways = $3^4$=81

But in these $3^4$ cases $D_4$ cases are onto

$\Rightarrow$ Required number of ways = 81 – $D_4$ = 81 – 9 =72

(S) Required number of function = $6.5.4.3-\left\{{}^4C_1.5.4.3-{}^4C_2.4,3-{}^4C_4.1\right\}=181$

3. Let L, M, N be the mid-points of the sides of $\Delta PQR$.




Area of $\Delta LMN=\frac{1}{2}|MN\times MN|=\frac{1}{2}|(-3b\hat{j}+3c\hat{k})\times(3a\hat{i}-3b\hat{j}|$

$=\frac{1}{2}|9(bc\hat{i}+ca\hat{j}+ab\hat{k}|=\frac{9}{2}\sqrt{(bc)^2+(ca)^2+(ab)^2}$

Now, area of $\Delta PQR=4\times$(Area of $\Delta LMN$)

$=4\times\frac{9}{2}\sqrt{b^2c^2+c^2a^2+a^2b^2}=18\sqrt{b^2c^2+c^2a^2+a^2b^2}$

4. We have, $w_i=lx_i+k$ for $i=1,2,....,n$



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$\Rightarrow \sum_{i=1}^{n} w_i = l\sum_{i=1}^{n} x_i + nk \Rightarrow \frac{1}{n}\sum_{i=1}^{n} w_i = l\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) + k \Rightarrow 55 = 48l + k$

Again, $w_i = lx_i + k$ for $i = 1, 2, \ldots n$

$\Rightarrow$ Var (W) = $l^2$ Var(X) $\Rightarrow Var(W) = l^2 Var(X) = \sigma_w = |l|\sigma_X$

$\Rightarrow 15 = 12|l| \Rightarrow |l| = \frac{5}{4} = 1.25 \Rightarrow l = \pm 1.25$

Substituting l = 1.25, and l = – 1.25 respectively, we get
K = – 5 and k = 115

5. Differentiating both sides of (a), we have $\frac{dy}{dx} = 3ae^{3x} + 5be^{5x} = 3y + 2Be^{5x}$

$\Rightarrow \frac{dy}{dx^2} = 3\frac{dy}{dx} + 10be^{5x} = 3\frac{dy}{dx} + 5\left(\frac{dy}{dx} - 3y\right) \Rightarrow \frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 15y = 0$

Differentiating both sides (b) we get $x\frac{dy}{dx} + y = ae^{x} - be^{-x} + 2x$

Again differentiating, we have $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = ae^{x} + be^{-x} + 2 = xy - x^2 + 2$

Hence the required differential equation is $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} - xy + x^2 - 2 = 0.$

Differentiating, (c) we have $2ax + 2By\frac{dy}{dx} = 0 \Rightarrow ax + by\frac{dy}{dx} = 0$ (1)

$\Rightarrow a + b\left(\frac{dy}{dx}\right)^2 + by\frac{d^2y}{dx^2} = 0$ (differentiating (1))

$\Rightarrow a + b\left[\left(\frac{dy}{dx}\right)^2 + y\frac{d^2y}{dx^2}\right] = 0$

Multiplying (2) by $x$ and subtracting from (1) we have

$bx\left[y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right] - by\frac{dy}{dx} = 0 \Rightarrow x\left\{y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2\right\} = y\frac{dy}{dx}$

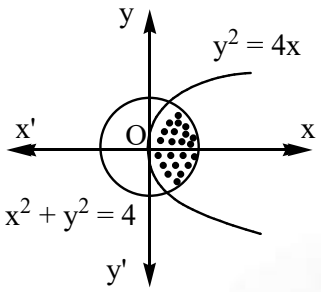
Given $y = a\sin(4x + b)$, we have $y_1 = 4a\cos(4x + b) \Rightarrow y_2 = -16a\sin(4x + b) = -16y.$

6. Since, $f : [0,1] \to [0,1]$ is a continuous function.
Therefore, Range $(f) \subset [0,1]$ and $f(x)$ attains every value between $f(0)$ and $f(1)$
Let $g(x) = f(x) - x$ for all $x \in [0,1]$.
Clearly, $g(x)$ is also continuous on [0, 1]
Also, $g(0) = f(0) - 0 = f(0) > 0$ and $g(1) = f(1) - 1 < 0$ $[\because f(1) < 1]$
Thus, $g(x)$ is continuous on [0, 1] such that $g(0)g(1) < 0$. Therefore, there exists a point $c \in (0,1)$ such that $g(c) = 0 \Rightarrow f(c) = c$
Hence, $f(x) = x$ for at least one $x \in (0,1)$.



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7. If the point $(-2a, a+1)$ lies in the shaded region in figure. Then,




$4a^2 + (a+1)^2 - 4 < 0$ and $(a+1)^2 - 4(-2a) < 0$

$\Rightarrow 5a^2 + 2a - 3 < 0$ and $a^2 + 10a + 1 < 0$

$\Rightarrow (a+1)(5a-3) < 0$ and $(a+5-2\sqrt{6})(a+5+2\sqrt{6}) < 0$

$\Rightarrow -1 < a < \frac{3}{5}$ and $-5-2\sqrt{6} < a < -52\sqrt{6}$

$\Rightarrow -1 < a < -5+2\sqrt{6} \Rightarrow a \in (-1, -5+2\sqrt{6}$

8. The plane passing through a and containing the line $r = b + \lambda c$ also passes through the point b and is parallel to the vector c. So, it is normal to the vector $(a - b) \times c$

Thus, the equation of the plane is $(r-a).|(a-b)\times c| = 0$

$\Rightarrow (r-a).(a-b)\times c = 0 \Rightarrow r.(a\times c - b\times c) = a.(a\times c - b\times c)$

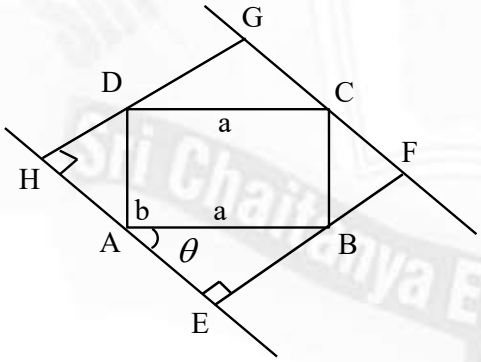
$\Rightarrow r.(a\times c - b\times c) = -a.(b\times c) \Rightarrow r.(b\times c + c\times a) - [abc] = 0$

$\because$ Length of the perpendicular form the origin to this plane

$$= \left| \frac{0.(b\times c + c\times a) - [abc]}{|b\times c + c\times a|} \right| = \frac{[abc]}{|b\times c + c\times a|}$$

9. Let ABCD be the given rectangle of sides a and b and EFGH be any rectangle, whose sides pass through A, B, C, D.

A = Area EFGH = (b sin $\theta$ + a cos $\theta$ ) (a sin $\theta$ + b cos $\theta$ ) = ab + ($a^2$ + $b^2$) sin $\theta$ cos $\theta$ .




$dA/d\theta = (a^2+b^2)\cos\theta$ so $dA/d\theta = 0 \Rightarrow \theta = \frac{\pi}{4}$

$$\frac{d^2A}{d\theta^2} = -2(a^2+b^2)\sin 2\theta, \text{ so } \left.\frac{d^2A}{d\theta^2}\right|_{\theta=\pi/4} < 0$$

Hence $A_{max} = \left(\frac{1}{2}\right)(a+b)^2$



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10. We rewrite function f(x) as $f(x)=\sqrt{a^2-1}\sin(ax-\phi)$ where $\phi=\arctan\frac{1}{a}$ its least Positive period is $\frac{2\pi}{a}$, and its amplitude is $\sqrt{a^2-1}$ .By symmetry of the figure enclosed by the graphs of the functions f(x) and g(x) . We can change the figure into a rectangle with $\sqrt{a^2-1}$ using the cut – and paste method

Therefore its area is $\frac{2\pi}{a}\sqrt{a^2-1}$

11. We have, $D=\begin{vmatrix}1 & \cos(\beta-\alpha) & \cos(\gamma-\alpha)\\ \cos(\alpha-\beta) & 1 & \cos(\gamma-\beta)\\ \cos(\alpha-\gamma) & \cos(\beta-\gamma) & 1\end{vmatrix}$

$$=\begin{vmatrix}\cos^2\alpha+\sin^2\alpha & \cos\beta\cos\alpha+\sin\beta\sin\alpha & \cos\gamma\cos\alpha+\sin\gamma\sin\alpha\\ \cos\alpha\cos\beta+\sin\alpha\sin\beta & \cos^2\beta+\sin^2\beta & \cos\gamma\cos\beta+\sin\gamma\sin\beta\\ \cos\alpha\cos\gamma+\sin\alpha\sin\gamma & \cos\beta\cos\gamma+\sin\beta\sin\gamma & \cos^2\gamma+\sin^2\gamma\end{vmatrix}$$

$$=\begin{vmatrix}\cos\alpha & \sin\alpha & 0\\ \cos\beta & \sin\beta & 0\\ \cos\gamma & \sin\gamma & 0\end{vmatrix}\begin{vmatrix}\cos\alpha & \sin\alpha & 0\\ \cos\beta & \sin\beta & 0\\ \cos\gamma & \sin\gamma & 0\end{vmatrix}=0$$

So, both the statement are true and statement 2 is a correct explanation for statement 1

12. $z^2-1=(z-1)(z-\alpha_1)(z-\alpha_2)+\ln(z-\alpha_4)$

$\log(z^5-1)=\ln(z-1)+\ln(z-\alpha_1+)\ln(z-\alpha_2)+\ln(z-\alpha_3)+\ln(z-\alpha_4)$

Differentiating $\frac{5z^4}{z^5-1}=\frac{1}{z-1}+\frac{1}{z-\alpha_1}+\frac{1}{z-\alpha_2}+\frac{1}{z-\alpha_3}+\frac{1}{z-\alpha_4}$

Putting $z=2\Rightarrow\sum_{r=1}^{4}\frac{1}{2-\alpha_1}=\frac{80}{31}-1=\frac{49}{31}$

13. Consider

$$\frac{\tan\theta^\circ}{1-3\tan^2\theta^\circ}=\frac{1}{8}\left[\frac{8\tan\alpha^\circ}{1-3\tan^2\theta^\circ}\right]+\tan\theta^\circ=\tan\theta^\circ=\frac{1}{8}(3\tan30^\circ-\tan\theta^\circ)$$

Put $\theta-8^\circ,24^\circ,72^\circ,216^\circ a$

Given expression =

$$\left(\frac{81}{8}\tan648^\circ-\frac{1}{8}\tan8^\circ\right)=\left(\frac{-81}{8}\tan72^\circ-\frac{1}{8}\tan8^\circ\right)=\left(\frac{81}{8}\tan108^\circ-\frac{1}{8}\tan8^\circ\right)\quad\therefore x+y=10$$

14. We have,

$$\lim_{x\to0}\frac{(1+x)^{1/x}-e+\frac{ex}{2}}{x^2}=\lim_{x\to0}\frac{e^{\frac{1}{x}\log(1+x)}-e+\frac{ex}{2}}{x^2}=\lim_{x\to0}\frac{e^{1-\frac{x}{2}+\frac{x^2}{3}-\frac{x^3}{4}+\ldots}-e+\frac{ex}{2}}{x^2}$$



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$$= e\lim_{x\to 0}\frac{e^{-\frac{x}{2}+\frac{x^2}{3}+\frac{x^3}{4}+\ldots}-1+\frac{x}{2}}{x^2} = e\lim_{x\to 0}\frac{1+\left(-\frac{x}{2}+\frac{x^2}{3}-\frac{x^3}{4}+\ldots\right)+\frac{1}{2!}\left(-\frac{x}{2}+\frac{x^2}{3}-\frac{x^3}{4}+\ldots\right)^2+\ldots.-1+\frac{x}{2}}{x^2}$$

$$= e\lim_{x\to 0}\frac{x^2\left(\frac{1}{3}+\frac{1}{2!}\times\frac{1}{4}\right)+x^3(\ldots.)+\ldots.}{x^2} = e\left(\frac{1}{3}+\frac{1}{8}\right)=\frac{11e}{24}$$

15. Find the sum of coefficients for terms without $x_4$

The sum of all coefficients in the expansion of $(x_1+x_2+x_3)^8$ is found by setting $x_1=x_2=x_3=1$, which gives $(1+1+1)^8=3^8$. This sum includes terms where $n_3=0$ and terms where $n_3\le 1$

Find the sum of coefficient for terms without $x_3$ and $x_4$

The terms that do not contains $x_3$ (and also not $x_4$) are those with $n_3 = 0$ and $n_4 = 0$.

These are terms in the expansion of $(x_1+x_2)^8$. The sum of their coefficients is found by setting $x_1=x_2=1$, which gives $(1+1)^8=2^8$

Calculate the final sum

The sum of coefficients for terms that contain $x_3$ but not $x_4$ is the total sum of coefficients for terms with out $x_4$ (from step 3) minus the sum of coefficients for terms without $x_3$ and $x_4$ (from step 4).

16. We know that $A.M. \ge G.M.$

$$\therefore \frac{27^{\cos x}+81^{\sin x}}{2} \ge \sqrt{27^{\cos x}\times 81^{\sin x}} \Rightarrow 27^{\cos x}+81^{\sin x} \ge 2\sqrt{3^{3\cos x+4\sin x}}$$

$$\Rightarrow 27^{\cos x}+81^{\sin x} \ge 2\sqrt{3^5}\,[\because -5\le 3\cos x+4\sin x\le 5] \Rightarrow 27^{\cos x}+81^{\sin x} \ge \frac{2}{9\sqrt{3}}$$

Hence, the minimum value of $27^{\cos x}+81^{\sin x}$ is $\frac{2}{9\sqrt{3}}$.

17. $$\int \frac{x^2-1}{x^2\left(x+\frac{1}{x}\right)\sqrt{x^2+\frac{1}{x^2}}}dx = \int \frac{1-\frac{1}{x^2}}{\left(x+\frac{1}{x}\right)\sqrt{\left(x+\frac{1}{x}\right)^2-2}}dx \text{ put } x+\frac{1}{x}=t$$

$$= \int \frac{dt}{t\sqrt{t^2-2}} \quad t=\sqrt{2}\sec\theta \qquad = \int \frac{\sqrt{2}\sec\theta\tan\theta d\theta}{\sqrt{2}\sec\theta\sqrt{2}\tan\theta} = \frac{1}{\sqrt{2}}\theta + C$$

18. We have,

$$\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{7}+\tan^{-1}\frac{1}{13}+\ldots.+\tan^{-1}\frac{1}{n^2+n+1}+\ldots.\text{to }\infty$$

$$= \lim_{n\to\infty}\sum_{r=1}^{n}\tan^{-1}\frac{1}{r^2+r+1} \qquad = \lim_{n\to\infty}\sum_{r=1}^{n}\tan^{-1}\left\{\frac{1}{1+r(r+1)}\right\}$$



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$= \lim_{n\to\infty}\sum_{r=1}^{n}\tan^{-1}\left\{\frac{(r+1)-r}{1+r(r+1)}\right\}$ $= \lim_{n\to\infty}\sum_{r=1}^{n}\left\{\tan^{-1}(r+1)-\tan^{-1}r\right\}$

$= \lim_{n\to\infty}\sum_{r=1}^{n}\left\{\tan^{-1}(n+1)-\tan^{-1}1\right\}$ $= \tan^{-1}\infty-\tan^{-1}1=\frac{\pi}{2}-\frac{\pi}{4}=\frac{\pi}{4}$

19. Foci of the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ is $(\pm1,0)$

Let the equation of the hyperbola be $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

Such that $2a-2\sin\theta\Rightarrow a=\sin\theta$ and $a^2+b^2=a^2e^2=1\Rightarrow b=\cos\theta$ So the required equation is

$\frac{x^2}{\sin^2\theta}-\frac{y^2}{\cos^2\theta}=1$ or $x^2\cos ec^2\theta-y^2\sec^2\theta=1$

20. Using $\int_0^a f(x)dx=\int_0^a f(a-x)dx$, we get

$I_1=\int_0^{\pi/2}f[\sin(\pi-2x)]\sin(\pi/2-x)dx$

$=\int_0^{\pi/2}f(\sin 2x)\cos x dx$ (2)

Adding (1) and (2) we get

$2I_1=\int_0^{\pi/2}f(\sin 2x)(\sin x+\cos x)dx$

$\sqrt{2}\int_0^{\pi/2}f(\sin 2x)\cos\left(x-\frac{\pi}{4}\right)dx$

Put $x-\pi/4=\theta$, so that

$2I_1=\sqrt{2}\int_{-\pi/4}^{\pi/4}f[\sin(\pi/2+2\theta)]\cos\theta d\theta$

$=\sqrt{2}\int_{-\pi/4}^{\pi/4}f(\cos 2\theta)\cos\theta d\theta$

$=2\sqrt{2}I_2$, as the integrand is an even function $\Rightarrow I_1/I_2=\sqrt{2}$.

21. Let $x=y=0\Rightarrow f(0)=1\, put x=y\Rightarrow 2f(x)\le\sin x+2\Rightarrow f(x)\le\sin x+1$

If $y=-x;1+f(0)\le f(x)+f(-x)\le 2$

$\Rightarrow f(-x)=2-f(x); f(-x)+f(-x)\le -2\sin x+2$

$\Rightarrow f(-x)\le-\sin x+1\Rightarrow 2-f(x)\le-\sin x+1\Rightarrow -f(x)\le-\sin x-1\Rightarrow f(x)\le 1+\sin x$

So, $f(x)=1+\sin x$

22. $29\left(\frac{31}{x+2}-7+\frac{31}{x+3}+8\right)=31\left(\frac{29}{x+1}-5+\frac{29}{x+4}-6\right)+370$

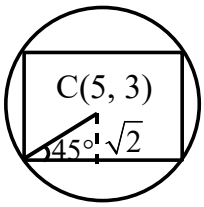
$29\times 31\left(\frac{1}{x+2}+\frac{1}{x+3}\right)=31\times 29\left(\frac{1}{x+1}+\frac{1}{x+4}\right)$

$\frac{1}{x+2}+\frac{1}{x+3}=\frac{1}{x+1}+\frac{1}{x+4}$ $\Rightarrow x=-5/2$ Ans: 5+2=7

23.



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One side of square is $y = x + k$

Distance of (5, 3) to the line $y = x + k$ is $\frac{|3-5-k|}{\sqrt{2}} = \sqrt{2} \Rightarrow |-2-k| = 2$

$\Rightarrow k = 0$ or $k = -4$

So lines are y =x and y = x – 4

Now, solving these lines with circle

y = x and $x^2 + y^2 - 10x - 6y + 30 = 0$

$\Rightarrow 2x^2 - 16x + 30 = 0 \Rightarrow x^2 - 8x + 15 = 0 \quad \Rightarrow x = 3,5 \Rightarrow x = 3, y = 3$

$x = 5, y = 5$ ; $y = x - 4$ *and* $x^2 + y^2 - 10x - 6y + 30 = 0$

$\Rightarrow x = 5, y = 1$ ; $\quad x = 7, y = 3$

Points (3, 3)(5, 5),(5, 1)(7, 3)

$\sum_{i=1}^{4} x_i^2 + y_i^2 = 9 + 9 + 25 + 25 + 25 + 1 + 49 + 9 = 152$

24. $n(A) = 100^3 - 80^3$, $n(B \cap A) = (100^3 - 60^3) - (80^3 - 40^3)$

↓ ↓

Minimum at most 40 & maximum upto 100 — Minimum [41, 80] maximum upto 80

Required probability = $\frac{100^3 - 60^3 - 80^3 + 40^3}{100^3 - 80^2} = \frac{42}{61} = \frac{p}{q} \Rightarrow p + q = 103$

25. BBB = I

ABBB = A

BAABB = A

BABAAB = A

$BABABA^2 = A$

$BABBA^2A^2 = A$

$BBAABA^2A^2 = A$

$BBABA^2A^2\ A^2 = A$

$BBBA^2A^2A^2A^2 = A$

$I.A^8 = A$

$A^8 = A$

$A^7 = I$ k = 7



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# PHYSICS

26. $\frac{1}{C_s}=\frac{1}{C}+\frac{1}{C}+\frac{1}{C}=\frac{3}{C}$

$C_s=\frac{C}{3}$ $\qquad V^1=V_1+V_2+V_3=V+V+V=3V$

27. Applying the lens maker formula

$\frac{1}{V}=\frac{1}{u}=\frac{1}{f}$ $\qquad m_T=\frac{v}{u}$

$u=-3.75 \quad f=+2.5 \qquad \vartheta=7.5cm$

Hence modulus of magnification of the objective lens is 2

28. $P\propto V^2$ $\qquad \frac{P}{P_0}=\left[\frac{V}{V_0}\right]^2 \qquad P=\left[\frac{V}{V_0}\right]^2 P_0$

29. Voltage across $200\Omega$ resistor is 50V so, equivalent resistance across voltmeter is also

$200\Omega \qquad 200=\frac{\frac{800}{3}.Rv}{\frac{800}{3}+Rv}\Rightarrow R=800\Omega$

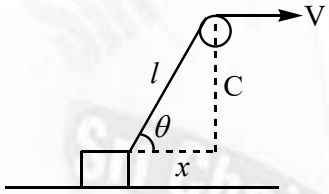
30. When the galvanometer shows zero deflection, hence Wheatstone bridge is balanced

$\frac{P}{Q}=\frac{R}{S}\Rightarrow\frac{10}{50}=\frac{\left(\frac{12x}{12+x}\right)}{20}\Rightarrow x=6\Omega$

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31. Static friction acting on the block $f_s=mg\sin 37°=2\times g\times\frac{3}{5}=12N$

32.




$l^2=x^2+c^2 \qquad 2l\frac{dl}{dt}=2x\frac{dx}{dt} \qquad \frac{dx}{dt}=\frac{lV}{x}=\frac{V}{\cos\theta}$

33. Apply loop laws and junction law calculate the currents.

34. On main scale 1cm is divided into 20 equal parts

Hence 1 M.S.D = $\frac{1}{20}$ cm.

Now least count = 1 M.S.D – 1 V.S.D = $\frac{M.S.D}{25}=\frac{1}{20\times 25}cm=0.002cm$

35. Voltmeter range = ($R_V$ + $R_A$) ammeter range

$6=(R_V+3)\,40\times10^{-3}=R_V+3=150=R_V=147$



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36. $F = Bqv\sin\theta$ If $\theta = 0(OR)\pi$ straight line.

If $\theta = \frac{\pi}{2}$ circle If $\theta = \frac{\pi}{3}$ Helix

37. As the mass number increases 2 to 56 binding energy per nucleon first increases maximum value at A = 56 and then decreases as mass number increases 56 to 235.

38. In given arrangement behaves as a AND gate.

39. $V_1 = \sqrt{\frac{T}{\mu}}$ $V_2 = \sqrt{\frac{2T}{\mu}}$ $\frac{V_1}{V_2} = \frac{1}{\sqrt{2}}$

40. Area of ellipse = $\pi ab$

Magnetic dipole moment = $\frac{\pi abi}{2}$

Torque $\overline{T} = \overline{M} \times \overline{B} = MB\sin 180° = 0$

41. At resonance $i_0 = \frac{E_0}{R}$ $i_0^1 = \frac{E_0}{R/2} = 2I_0$

42. $T^2 \propto R^3$ for circular orbits. Hence reason is correct explanation of assertion.

43. Conservative forces changes the K.E but not changes the M.E.

44. $E = \frac{hc}{\lambda} \Rightarrow h = ML^2T^{-1}$ $B = \frac{\Delta P}{\left(\frac{-\Delta V}{V}\right)} = ML^{-1}T^{-2}$

45. Gases has less viscosity due to insoluble impurities like detergent surface tension decreases.

46. Case I : $\phi = \frac{q}{24 \in_0}$ Case II : $k\phi = \frac{q}{6 \in_0} \Rightarrow k = 4$

47. $u\delta =$ Constant

$p =$ Constant $\frac{du}{d\omega} = \left(\frac{C_V}{C_P - C_V}\right) = \frac{3}{2}$

48. $\frac{1}{2}kA^2 = 1J$ $k = \frac{2}{A^2} = \frac{2}{(0.4)} = \frac{25}{2}$

$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{2}{25/2}} = \frac{4\pi}{5}$

49. $d = f_1 - |f_2| = 30 - 10 = 20\text{cm}$

50. Work done by static frictions is zero Hence mechanical energy is constant

$\frac{1}{2}kx^2 + \frac{1}{2}mv_{cm}^2(1+B) =$ Constant

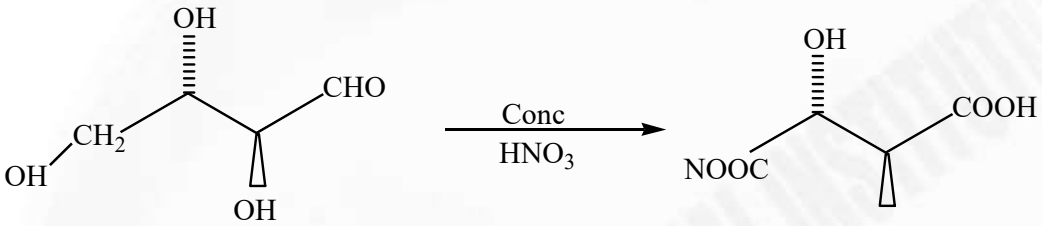
$T = \pi\sqrt{6M/k}$ $T = 2\pi\sqrt{\frac{M(1+B)}{K}}$

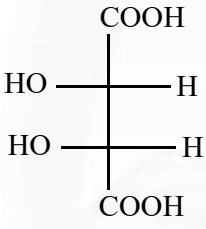


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# CHEMISTRY

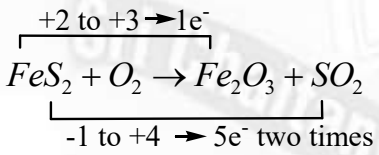
51. Resonance energy makes benzene more stable
52. Amide gets hydrolysed in acidic medium, halide should undergo elimination in alkaline medium
53. Anilinium ion is electronically more deficient
54. If the difference in boiling point is more then simple distillation is used
55. IUPAC rules
56. $-OCH_3$ has +M effect
57. Structure given by NCERT
58.



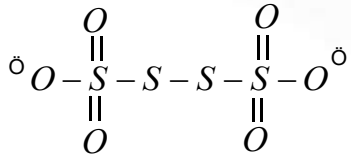




meso compound

59. Solubility increases with pressure, Henry's Law
60. $q_{rev} > q_{irrev}$ in a isobaric change so entropy also differs
61. $E = 0.059\ (P^H_{cathode} - P^H_{anode})$
$= 0.059\times(9-7)$
$= 2\times0.059$
$= 0.118$ volt
$E_{cell}$ is positive cell will work spontaneous
62.




63. $Na_2S_2O_3 + I_2 \rightarrow S_4O_6^{2-} + 2I^-$



Three S – S linkages

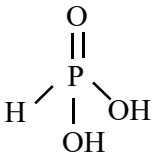
64. Half life does not change on varying conc of A hence it is 1st order w.r.t A and half life doubles when concentration of B doubles hence it is zero$^{th}$ order w.r.t B. Overall order is 1.



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65. Le Chatlier's Principle

66. $PCl_3 + 3H_2O \rightarrow P(OH)_3 + 3HCl$



= 6 moles for $H_3PO_3$
= 3 moles for HCl
Total 9 moles of $OH^-$ required

67. $NH_4Cl + NaNO_2 \rightarrow NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

68. $I_3^{-1}, Cr_2O_7^{2-}$ *and* $CH_3 - OCH_3$ have bond angle greater than 109°28'

69. All electrons are paired hence less M.P in 3d series

70. All shapes according to VSEPR theory

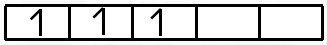
71. $[M(AA)_3]$ shows optical activity

72. $3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$

73. $BrF_3, PCl_5, SF_4$ has $Sp^3d$ hybridization

74. Berrylium acetate (alk), $H_2S_2O_7$, $N_2O_3$(symm)$Mn_2O_7$ has E – O – E link.

75. $Cr^{3+} \Rightarrow 4S^\circ 3d^3$




Spin multiplicity = 2S + 1 = 2 × 3/2 + 1 = 4

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